

lenguaje SQL II

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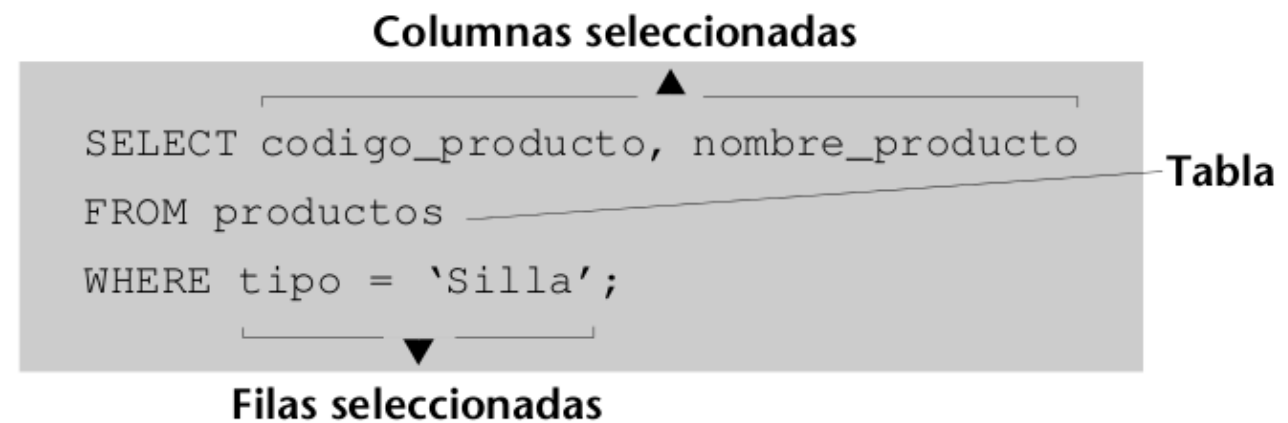
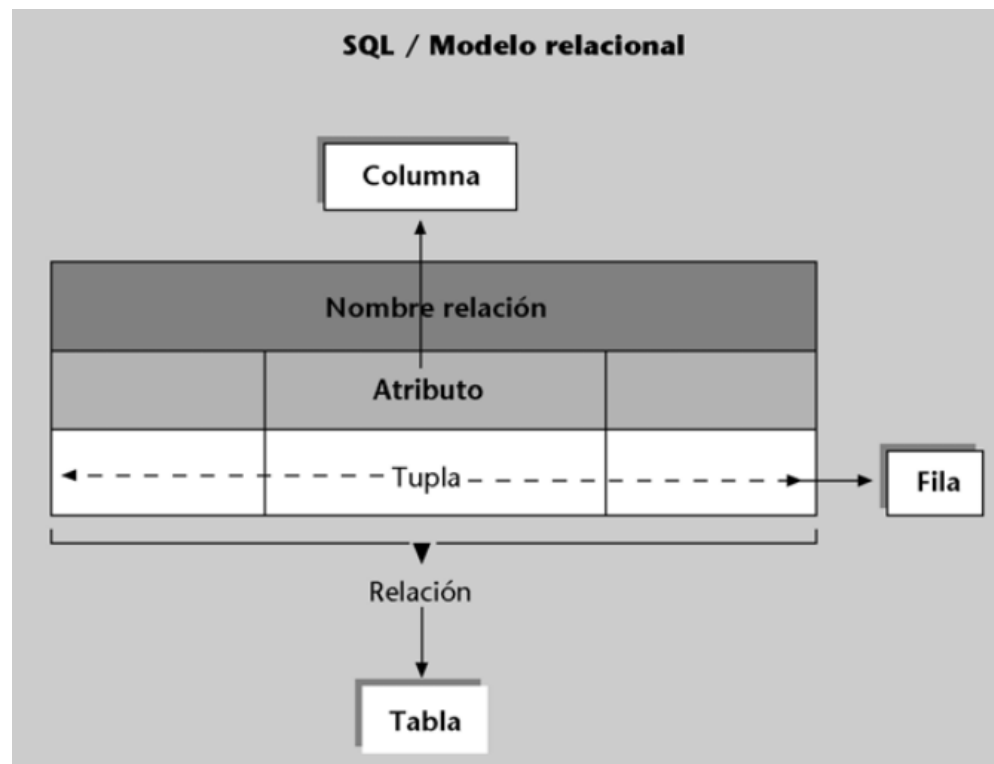
Contenidos

- Repaso
- Algebra relacional
- SQL II

Qué es el SQL?

Introducción

- Lenguaje estándar ANSI(1986) de **definición, manipulación y control** de bases de datos relacionales.
- Estándar de facto en la actualidad, aunque cada fabricante agrega sus propias extensiones.



SQL

DDL- Data Definition Language

- **DDL (lenguaje de definición de datos)**
 - Las sentencias DDL son aquellas utilizadas para la creacion de una base de datos y todos sus componentes.
 - Tablas, indices, relaciones, restricciones...
- **Sentencias**
 - CREATE -> crear tablas y vistas.
 - DROP -> para borrar tablas y vistas.
 - ALTER -> para modificar tablas.
 - ALTER TABLE EMPLEADO ADD COLUMN sueldo INT;
 - ALTER TABLE EMPLEADO DROP COLUMN mail CASCADE;
 - ALTER TABLE DEPARTAMENTO ALTER COLUMN id_director SET DEFAULT '00000000';

SQL

Sentencia CREATE

Restricciones de columna	
Restricción	Descripción
NOT NULL	La columna no puede tener valores nulos.
UNIQUE	La columna no puede tener valores repetidos. Es una clave alternativa.
PRIMARY KEY	La columna no puede tener valores repetidos ni nulos. Es la clave primaria.
REFERENCES tabla [(columna)]	La columna es la clave foránea de la columna de la tabla especificada.
CHECK (condiciones)	La columna debe cumplir las condiciones especificadas.

```
FOREIGN KEY clave_secundaria REFERENCES tabla [(clave_primaria)]  
[ON DELETE {NO ACTION | CASCADE | SET DEFAULT | SET NULL}]  
[ON UPDATE {NO ACTION | CASCADE | SET DEFAULT | SET NULL}]
```



```
FOREIGN KEY(trackartist) REFERENCES artist(artistid)
```

Restricciones de tabla	
Restricción	Descripción
PRIMARY KEY (columna [, columna...])	El conjunto de las columnas especificadas no puede tener valores nulos ni repetidos. Es una clave primaria.
FOREIGN KEY (columna [, columna...]) REFERENCES tabla [(columna2 [, columna2...])]	El conjunto de las columnas especificadas es una clave foránea que referencia la clave primaria formada por el conjunto de las columnas2 de la tabla dada. Si las columnas y las columnas2 se denominan exactamente igual, entonces no sería necesario poner columnas2.
CHECK (condiciones)	La tabla debe cumplir las condiciones especificadas.

SQL

DML- Data Manipulation Language

- DML (Lenguaje de manipulación de datos):
 - Utilizadas para **INSERT**, borrar **DELETE**, modificar **UPDATE** y consultar **SELECT FROM** los datos de una base de datos.

```
SELECT nombre_columnas_a_seleccionar  
FROM tabla_a_consultar  
WHERE condiciones;
```

SQL

SELECT

```
SELECT nombre_columnas_a_seleccionar  
FROM tabla_a_consultar  
WHERE condiciones;
```

- select * from Jugador;
- select nombre, edad from Jugador;
- select nombre from Jugador where edad > 30;
- select nombre from Jugador where edad > 30 or id_j < 10;

SQL

Funciones de agregación

- **COUNT**

- Select count(id_j) from Jugador where edad > 31;

- **SUM**

- select sum(edad) from Jugador where edad > 31;

- **AVG**

- select avg(edad) from Jugador where edad > 31;

- **MIN y MAX**

- select max(edad) from Jugador where edad >31;

SQL

BETWEEN y IN

- Select * from Jugador where edad **between** 20 and 30;
- Select * from Jugador where id_j **in** (1,20,35);
- Select * from Jugador where id_j **not in** (1,20,35);

Algebra relacional

Algebra relacional

Operadores

- Operaciones fundamentales para recuperar y manipular tuplas en una relación.
 - Basado en algebra de conjuntos.
- Cada operador toma uno o más relaciones como entradas y su salida es una nueva relacion.
 - Podemos "encadenar" operadores para crear operaciones más complejas.
- Operadores:
 - σ [select], $\cup$ [union], $\cap$ [intersection], $-$ [Difference], $\times$ [Product], $\bowtie$ [join]

Algebra relacional

Ejemplos select

- Seleccionar un subconjunto de las tuplas de una relación que satisfice unos criterios de seleccion (predicado)
- El predicado actúa como un filtro para seleccionar solo tuplas que cumplen su condición.
- Podemos combinar múltiples predicados (AND, OR..)

R(a_id, b)

a_id	b
a1	10
a2	20
a2	31
a4	32

$$\sigma_{a_id='a1'}(R) \quad \sigma_{a_id='a2' \wedge b > 30}(R)$$

a_id	b
a1	10

a_id	b
a2	31

```
SELECT * FROM R
WHERE a_id = 'a2' and B > 30
```

Algebra relacional

UNION, INTERSECTION, DIFFERENCE, PRODUCT

```
1 %%sql
2 drop table if exists R;
3 drop table if exists S;
4 -- Relación R --
5 CREATE TABLE R (
6 a_id INTEGER,
7 b INTEGER);
8 -- Relación S --
9 CREATE TABLE S (
10 a_id INTEGER,
11 b INTEGER);
12
13 -- valores en R --
14 INSERT INTO R(a_id,b) VALUES
15 (1,10),
16 (2,20),
17 (2,31),
18 (4,32);
19 -- valores en S --
20 INSERT INTO S(a_id,b) VALUES
21 (4,32),
22 (5,50),
23 (7,51);
```

$R \cup S$

```
1 %%sql
2 select * from R
3 UNION ALL
4 select * from S;
```

* sqlite:///test.db
Done.

a_id	b
1	10
2	20
2	31
4	32
4	32
5	50
7	51

$R \cap S$

```
1 %%sql
2 select * from R
3 INTERSECT
4 select * from S;
```

* sqlite:///test.db
Done.

a_id	b
4	32

$R \times S$

```
1 %%sql
2 select * from R,S;
```

* sqlite:///test.db
Done.

a_id	b	a_id_1	b_1
1	10	4	32
1	10	5	50
1	10	7	51
2	20	4	32
2	20	5	50
2	20	7	51
2	31	4	32
2	31	5	50
2	31	7	51
4	32	4	32
4	32	5	50
4	32	7	51

$R - S$

```
1 %%sql
2 select * from R
3 EXCEPT
4 select * from S;
```

* sqlite:///test.db
Done.

a_id	b
1	10
2	20
2	31

Algebra relacional

JOIN

```
1 %%sql
2 drop table if exists R;
3 drop table if exists S;
4 -- Relación R --
5 CREATE TABLE R (
6 a_id INTEGER,
7 b INTEGER);
8 -- Relación S --
9 CREATE TABLE S (
10 a_id INTEGER,
11 b INTEGER);
12
13 -- valores en R --
14 INSERT INTO R(a_id,b) VALUES
15 (1,10),
16 (2,20),
17 (2,31),
18 (4,32);
19 -- valores en S --
20 INSERT INTO S(a_id,b) VALUES
21 (4,32),
22 (5,50),
23 (7,51);
```

Generar una relación que contenga todas
Las tuplas que son una combinación de dos
tuplas (una de cada relación de entrada)
con un valor(es) común(es) para uno o
más atributos.

$$R \bowtie S$$

```
1 %%sql
2 select * from R NATURAL JOIN S;
```

* sqlite:///test.db
Done.
a_id b
4 32

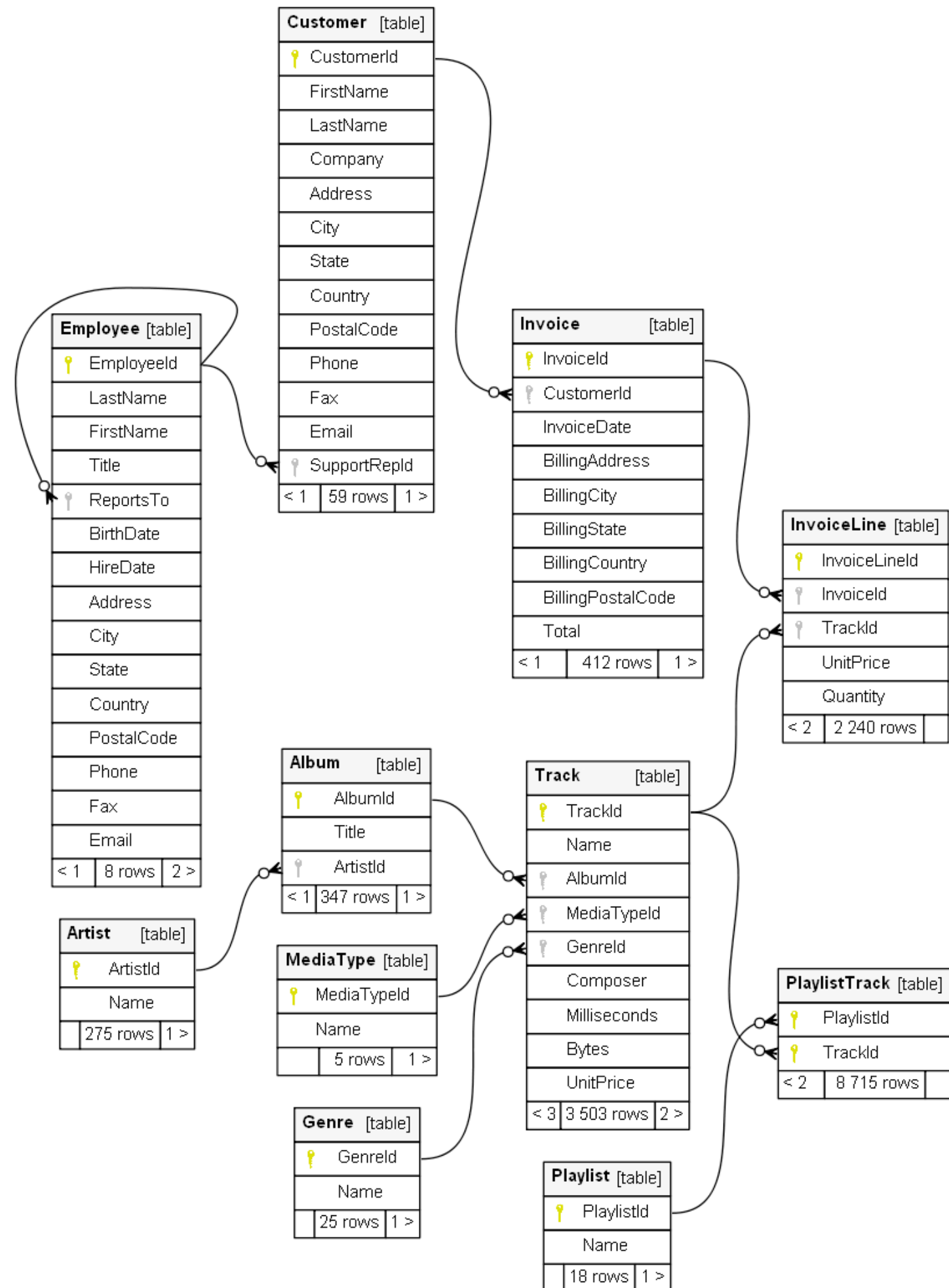
- RENAME
- SORTING
- AGGREGATION

SQL II

SQL II

Consultas combinando 2 o más relaciones

- Base de datos **Chinook**
- La base de datos Chinook modela una tienda de medios digitales, que incluye tablas para artistas, álbumes, canciones, facturas y clientes.



SQL II

as, DISTINCT, GROUP BY, and HAVING

- Contar el número de canciones?

```
select COUNT(TrackID) as cnt from Track;  
cnt 3503
```

- Mostrar las listas distintas desde PlaylistTrack?

```
select DISTINCT PlaylistId from PlaylistTrack;  
  
select COUNT(DISTINCT PlaylistId) as cnt from PlaylistTrack;  
cnt 14
```

- Contar el número de canciones por lista?

```
select PlaylistId, COUNT(TrackId) as cnt  
from PlaylistTrack GROUP BY PlaylistId;
```

- Seleccionar las listas con más de 100 canciones?

```
select PlaylistId, COUNT(TrackId) as cnt  
from PlaylistTrack  
GROUP BY PlaylistId  
HAVING cnt > 100;
```

PlaylistId	cnt	PlaylistId	cnt
1	3290	1	3290
3	213	3	213
5	1477	5	1477
8	3290	8	3290
9	1	10	213
10	213		
11	39		
12	75		
13	25		
14	25		
15	25		

SQL II

Combinando relaciones

- Seleccionar el nombre del artista y el nombre del album

```
select Name, Title as album_name
from album as al, Artist as a
where al.Artistid = a.Artistid limit 10;
```

- Mostrar los album de AC/DC

```
select Name, Title as album_name
from album as al, Artist as a
where al.Artistid = a.Artistid and a.Name = "AC/DC";
```

Name	album_name
AC/DC	For Those About To Rock We Salute You
Accept	Balls to the Wall
Accept	Restless and Wild
AC/DC	Let There Be Rock
Aerosmith	Big Ones
Alanis Morissette	Jagged Little Pill
Alice In Chains	Facelift
Antônio Carlos Jobim	Warner 25 Anos
Apocalyptica	Plays Metallica By Four Cellos
Audioslave	Audioslave

Name	album_name
AC/DC	For Those About To Rock We Salute You
AC/DC	Let There Be Rock

Operaciones con strings

LIKE

- **LIKE** es usado para realizar búsquedas en texto.
- Operadores:
 - ‘%’ -> acepta cualquier substring.

```
select distinct(composer) from track where composer like  
"%Brian";  
select distinct(composer) from track where composer like  
"%Brian";  
select distinct(composer) from track where composer like  
"%Brian%";
```

- ‘_’ -> cualquier caracter

```
select distinct(composer) from track where composer like  
"%Br_an%";
```

```
1 %%sql  
2 select distinct(composer) from track where co  
* sqlite:///content/chinook-database/ChinookDa  
Done.  
Composer  
May, Brian  
Deacon, John/May, Brian
```

```
[ 86] 1 %%sql  
2 select distinct(composer) from track where co  
* sqlite:///content/chinook-database/ChinookDa  
Done.  
Composer  
Brian Holland/Freddie Gorman/Georgia Dobbins/Robert Batemar  
Brian May  
Brian Holland/Eddie Holland/Lamont Dozier  
Brian Eno, Bono, Adam Clayton, The Edge & Larry Mullen Jnr.  
Brian Eno/U2
```

```
[ 84] 1 %%sql  
2 select distinct(composer) from track where co  
* sqlite:///content/chinook-database/ChinookDa  
Done.  
Composer  
Angus Young, Malcolm Young, Brian Johnson  
Brian Holland/Freddie Gorman/Georgia Dobbins/Robert Batemar  
Brian May  
Brian Holland/Eddie Holland/Lamont Dozier  
Brian Eno, Bono, Adam Clayton, The Edge & Larry Mullen Jnr.  
May, Brian  
Deacon, John/May, Brian  
Brian Eno/U2
```

Operaciones con strings

Otras funciones

- Length, upper, lower, substr, replace, ||

```
select length(distinct(composer)) from track where  
composer like "%Br_an%" limit 2;
```

```
select upper(distinct(composer)) from track where  
composer like "%Br_an%" limit 2;
```

```
select substr(distinct(composer),1,6) from track  
where composer like "%Br_an%" limit 2;
```

```
select replace(distinct(composer),"Angus","Alex")  
from track where composer like "%Br_an%" limit 2;
```

```
select composer || Name as merge_composer_name from  
track where composer like "%Br_an%" limit 2;
```

```
1 %%sql  
2 select length(distinct(composer)) from track where composer like  
  
* sqlite:///content/chinook-database/ChinookDatabase/DataSources/  
Done.  
length(distinct(composer))  
41  
41
```

```
1 %%sql  
2 select upper(distinct(composer)) from track where composer like  
  
* sqlite:///content/chinook-database/ChinookDatabase/DataSources/  
Done.  
upper(distinct(composer))  
ANGUS YOUNG, MALCOLM YOUNG, BRIAN JOHNSON  
ANGUS YOUNG, MALCOLM YOUNG, BRIAN JOHNSON
```

```
1 %%sql  
2 select substr(distinct(composer),1,6) from track where composer  
  
* sqlite:///content/chinook-database/ChinookDatabase/DataSources/  
Done.  
substr(distinct(composer),1,6)  
Angus  
Angus
```

```
1 %%sql  
2 select replace(distinct(composer),"Angus","Alex") from track whe  
  
* sqlite:///content/chinook-database/ChinookDatabase/DataSources/  
Done.  
replace(distinct(composer),"Angus","Alex")  
Alex Young, Malcolm Young, Brian Johnson  
Alex Young, Malcolm Young, Brian Johnson
```

```
1 %%sql  
2 select composer || Name as merge_composer_name from track where  
  
* sqlite:///content/chinook-database/ChinookDatabase/DataSources/  
Done.  
merge_composer_name  
Angus Young, Malcolm Young, Brian JohnsonFor Those About To Rock (We Salute You)  
Angus Young, Malcolm Young, Brian JohnsonPut The Finger On You
```

Controlando los resultados

ORDER BY, LIMIT

- ORDER BY <atributo> [ASC|DESC]
- Ordena las tuplas por uno o más valores de sus columnas

```
select Name,Milliseconds from track
order by Milliseconds ASC limit 3;
```

```
select Name,Milliseconds from track
order by Milliseconds DESC limit 3;
```

```
select * from track
order by MediaTypeId DESC, GenreId ASC, AlbumId DESC
limit 10;
```

- LIMIT X [offset] Y

```
select * from track limit 5 offset 10
```

```
1 %%sql
2 select Name,Milliseconds from track
3 order by Milliseconds ASC limit 3;
```

* sqlite:///content/chinook-database/ChinookDatabase/DataSources.
Done.

Name	Milliseconds
É Uma Partida De Futebol	1071
Now Sports	4884
A Statistic	6373

```
1 %%sql
2 select Name,Milliseconds from track
3 order by Milliseconds DESC limit 3;
```

* sqlite:///content/chinook-database/ChinookDatabase/DataSources.
Done.

Name	Milliseconds
Occupation / Precipice	5286953
Through a Looking Glass	5088838
Greetings from Earth, Pt. 1	2960293

```
1 %%sql
2 select * from track
3 order by MediaTypeId DESC, GenreId ASC, AlbumId DESC limit 10;
```

* sqlite:///content/chinook-database/ChinookDatabase/DataSources.
Done.

TrackId	Name	AlbumId	MediaTypeId	GenreId	
3355	Love Comes	265	5	1	Darius "Take One" Min
3353	I Guess You're Right	265	5	1	Darius "Take One" Min
3357	OAM's Blues	267	5	2	Aaron Goldberg
3350	Despertar	262	5	2	Andrea Dulbecco
3349	Amanda	262	5	2	Luca Gusella
3356	Muita Bobeira	266	5	7	Luciana Souza
3358	One Step Beyond	264	5	15	Karsh Kale
3352	Distance	264	5	15	Karsh Kale/Vishal Vaid
3354	I Ka Barra (Your Work)	263	5	16	Habib Koité
3351	Din Din Wo (Little Child)	263	5	16	Habib Koité

```
1 %%sql
2 select * from track limit 5 offset 10
```

* sqlite:///content/chinook-database/ChinookDatabase/DataSources.
Done.

TrackId	Name	AlbumId	MediaTypeId	GenreId	Con
11	C.O.D.	1	1	1	Angus Young, Malcoli
12	Breaking The Rules	1	1	1	Angus Young, Malcoli
13	Night Of The Long Knives	1	1	1	Angus Young, Malcoli
14	Spellbound	1	1	1	Angus Young, Malcoli
15	Go Down	4	1	1	AC/DC

Manipulando fechas y horas

strftime

- Atributos tipo DATETIME
 - TEXTO “YYYY-MM-DD HH:MM:SS.SSS”
- Función strftime permite formatear fechas y realizar calculos con fechas.

Format	Explanation
%d	Day of the month (1-31)
%f	Seconds with fractional seconds (SS.sss)
%H	Hour on 24-hour clock (00-23)
%j	Day of the year (001-366)
%J	Julian day number (DDDDDDD.dddddd)
%m	Month (01-12)
%M	Minute (00-59)
%s	Seconds since 1970-01-01
%S	Seconds (00-59)
%w	Weekday (0-6) (0=Sunday, 1=Monday, 2=Tuesday, 3=Wednesday, 4=Thursday, 5=Friday, 6=Saturday)
%W	Week number in the year (00-53) The first Monday is the beginning of week 1.
%Y	Year with century (yyyy)
%%	% as a literal

Manipulando fechas y horas

Strftime

- Atributos tipo DATETIME
- Función strftime permite retornar fechas con formato

```
PRAGMA table_info([invoice]);
```

```
select InvoiceDate,  
strftime('%Y',InvoiceDate) as "Año",  
strftime('%m',InvoiceDate) as "Mes",  
strftime('%d',InvoiceDate) as "Dia",  
strftime('%H',InvoiceDate) as "Horas",  
strftime('%M',InvoiceDate) as "Minutos"  
from invoice limit 5;
```

```
select InvoiceDate,  
strftime('%Y',InvoiceDate) as "Año",  
strftime('%m',InvoiceDate) as "Mes",  
strftime('%d',InvoiceDate) as Dia  
from invoice  
where Dia in ("01","02","11") limit 5;
```

```
1 %%sql  
2 PRAGMA table_info([invoice]);
```

```
* sqlite:///content/chinook-database/ChinookDatabase/DataSo  
Done.
```

cid	name	type	notnull	dflt_value	pk
0	InvoiceId	INTEGER	1	None	1
1	CustomerId	INTEGER	1	None	0
2	InvoiceDate	DATETIME	1	None	0
3	BillingAddress	NVARCHAR(70)	0	None	0
4	BillingCity	NVARCHAR(40)	0	None	0
5	BillingState	NVARCHAR(40)	0	None	0
6	BillingCountry	NVARCHAR(40)	0	None	0
7	BillingPostalCode	NVARCHAR(10)	0	None	0
8	Total	NUMERIC(10,2)	1	None	0

```
1 %%sql  
2 select InvoiceDate,strftime('%Y',InvoiceDate) as "Año",  
3 strftime('%m',InvoiceDate) as "Mes",  
4 strftime('%d',InvoiceDate) as "Dia",  
5 strftime('%H',InvoiceDate) as "Horas",  
6 strftime('%M',InvoiceDate) as "Minutos"  
7 | from invoice limit 5;
```

```
* sqlite:///content/chinook-database/ChinookDatabase/DataSo  
Done.
```

InvoiceDate	Año	Mes	Dia	Horas	Minutos
2009-01-01 00:00:00	2009	01	01	00	00
2009-01-02 00:00:00	2009	01	02	00	00
2009-01-03 00:00:00	2009	01	03	00	00
2009-01-06 00:00:00	2009	01	06	00	00
2009-01-11 00:00:00	2009	01	11	00	00

```
1 %%sql  
2 select InvoiceDate,strftime('%Y',InvoiceDate) as "Año",  
3 strftime('%m',InvoiceDate) as "Mes",  
4 strftime('%d',InvoiceDate) as Dia  
5 | from invoice where Dia in ("01","02","11") limit 5;
```

```
* sqlite:///content/chinook-database/ChinookDatabase/DataSo  
Done.
```

InvoiceDate	Año	Mes	Dia
2009-01-01 00:00:00	2009	01	01
2009-01-02 00:00:00	2009	01	02
2009-01-11 00:00:00	2009	01	11
2009-02-01 00:00:00	2009	02	01
2009-02-01 00:00:00	2009	02	01

Almacenando resultados en tablas

CREATE TABLE from SELECT

- Almacenar resultados de consultas en tablas
- Tablas no deben estar definidas.
- Tablas deben tener el mismo numero y tipo de columnas

```
create table Fecha as
select strftime('%Y',InvoiceDate) as "Año",
strftime('%m',InvoiceDate) as "Mes",
strftime('%d',InvoiceDate) as "Dia"
from invoice limit 10;
```

```
CREATE TABLE new_table AS
SELECT expressions
FROM existing_tables
[WHERE conditions];
```

```
1 %%sql
2 select strftime('%Y',InvoiceDate) as "Año",
3 strftime('%m',InvoiceDate) as "Mes",
4 strftime('%d',InvoiceDate) as "Dia"
5 | from invoice limit 5;
```

* sqlite:///content/chinook-database/ChinookDatabase.db
Done.

Año	Mes	Dia
2009	01	01
2009	01	02
2009	01	03
2009	01	06
2009	01	11

```
1 %%sql
2 create table Fecha as
3 select strftime('%Y',InvoiceDate) as "Año",
4 strftime('%m',InvoiceDate) as "Mes",
5 strftime('%d',InvoiceDate) as "Dia"
6 | from invoice limit 10;
```

* sqlite:///content/chinook-database/ChinookDatabase.db
Done.

[]

```
1 %%sql
2 select * from Fecha limit 10;
```

* sqlite:///content/chinook-database/ChinookDatabase.db
Done.

Año	Mes	Dia
2009	01	01
2009	01	02
2009	01	03
2009	01	06
2009	01	11
2009	01	19
2009	02	01

Subconsultas en SQL

SELECT (SELECT)

- Consultas conteniendo subconsultas
- Subconsultas pueden ser usadas casi en cualquier lugar de las consultas.
- Select .. from where x IN (select ...)

```
select name , AlbumId
from track
where AlbumId in (
    select AlbumId
    from album
    where title = "Californication")
limit 10;
```

```
1 #Obtener los nombres de todas las canciones del álbum "Californication".
2 %%sql
3 select name , AlbumId
4 from track
5 where AlbumId in (
6     select AlbumId
7     from album
8     where title = "Californication")
9 limit 5;
```

* sqlite:///content/chinook-database/ChinookDatabase/DataSources/Chinook_S
Done.

Name	AlbumId
Around The World	195
Parallel Universe	195
Scar Tissue	195
Otherside	195
Get On Top	195

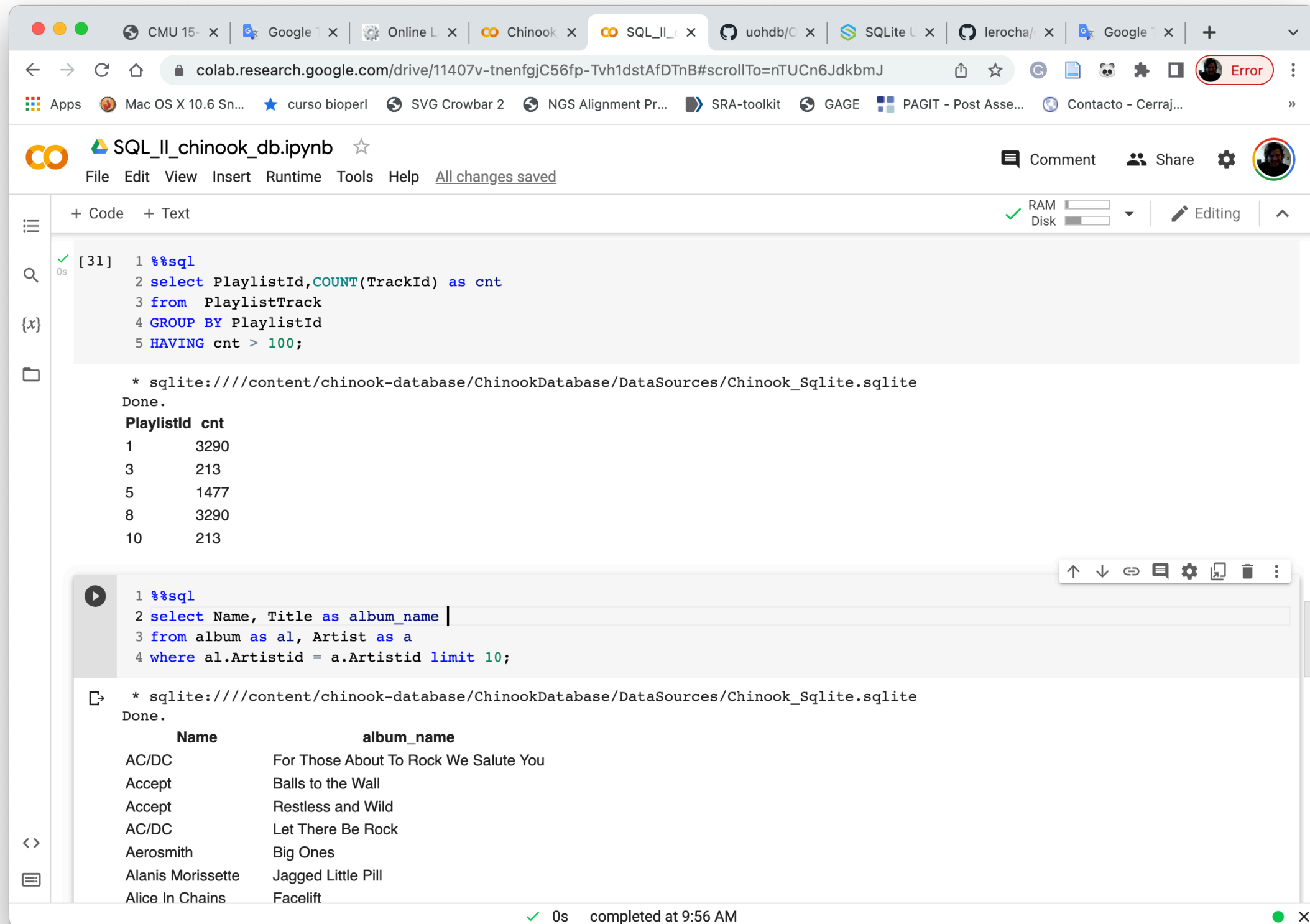
```
1 %%sql
2 select name , AlbumId
3 from track
4 where AlbumId not in (
5     select AlbumId
6     from album
7     where title = "Californication")
8 limit 5;
```

* sqlite:///content/chinook-database/ChinookDatabase/DataSources/Chinook_S
Done.

Name	AlbumId
For Those About To Rock (We Salute You)	1
Balls to the Wall	2
Fast As a Shark	3
Restless and Wild	3
Princess of the Dawn	3

Practicar SQL

Google Colab



The screenshot shows a Google Colab notebook titled "SQL_II_chinook_db.ipynb". The notebook has two code cells. The first cell contains a SQL query that counts the number of tracks per playlist. The second cell contains a SQL query that lists the top 10 albums by artist. Both queries are executed successfully, and the results are displayed below each query.

Code Cell 1:

```
1 %%sql
2 select PlaylistId, COUNT(TrackId) as cnt
3 from PlaylistTrack
4 GROUP BY PlaylistId
5 HAVING cnt > 100;
```

Output 1:

```
* sqlite:///content/chinook-database/ChinookDatabase/DataSources/Chinook_Sqlite.sqlite
Done.
PlaylistId cnt
1          3290
3          213
5          1477
8          3290
10         213
```

Code Cell 2:

```
1 %%sql
2 select Name, Title as album_name
3 from album as al, Artist as a
4 where al.Artistid = a.Artistid limit 10;
```

Output 2:

```
* sqlite:///content/chinook-database/ChinookDatabase/DataSources/Chinook_Sqlite.sqlite
Done.
Name          album_name
AC/DC          For Those About To Rock We Salute You
Accept         Balls to the Wall
Accept         Restless and Wild
AC/DC          Let There Be Rock
Aerosmith      Big Ones
Alanis Morissette Jagged Little Pill
Alice In Chains Facelift
```

The bottom status bar indicates that the execution was successful (green checkmark) and completed at 9:56 AM.

Consultas?

Consultas o comentarios?

Muchas gracias